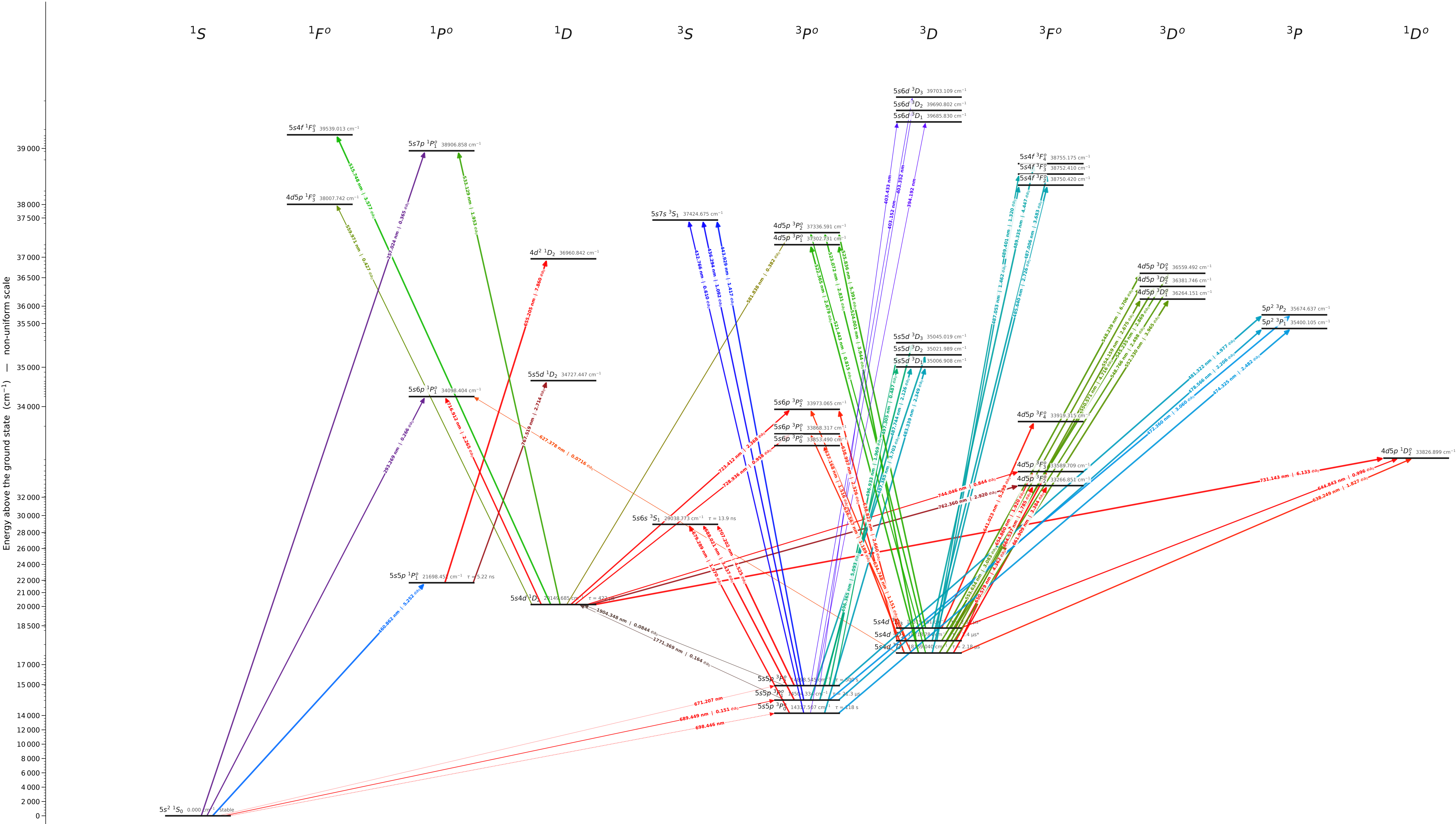
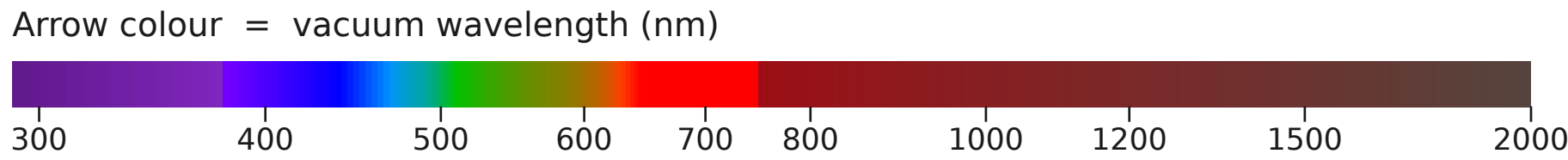


# Strontium-86

<sup>86</sup>Sr (Sr I), nuclear spin *I* = 0. Energy levels and transitions below 2 μm: 40 levels, 70 lines.

Sr<sup>+</sup> ionisation limit 45932.20 cm<sup>-1</sup> = 5.69487 eV (217.712 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element |(U||*er*||*J'*)| in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, *n* = 70.  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



## Key transitions

| Transition                                                                                  | λ vac (nm) | λ air (nm) | ν (THz)  | (U   <i>er</i>    <i>J'</i> )  (ea <sub>0</sub> ) | A (s <sup>-1</sup> ) | Γ/2π up. (MHz) | branch (%) | Use                                                   |
|---------------------------------------------------------------------------------------------|------------|------------|----------|---------------------------------------------------|----------------------|----------------|------------|-------------------------------------------------------|
| 5s5p <sup>3</sup> P <sub>0</sub> – 5s6d <sup>3</sup> D <sub>1</sub>                         | 394.1924   | 394.0808   | 760.5232 | –                                                 | –                    | –              | –          | 394 nm                                                |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>3</sub>                         | 403.1516   | 403.0377   | 743.6221 | –                                                 | –                    | –              | –          | 403.15 nm                                             |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>2</sub>                         | 403.3517   | 403.2378   | 743.2532 | –                                                 | –                    | –              | –          | 403.35 nm repumper                                    |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>1</sub>                         | 403.4326   | 403.3187   | 743.1041 | –                                                 | –                    | –              | –          | 403.43 nm                                             |
| 5s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s5p <sup>1</sup> P <sub>1</sub> <sup>o</sup> | 460.8624   | 460.7333   | 650.5032 | 5.252                                             | 1.903e+08            | 30.51          | 100        | 461 nm blue MOT / Zeeman slower / imaging             |
| 5s5p <sup>3</sup> P <sub>0</sub> – 5p <sup>2</sup> <sup>3</sup> P <sub>1</sub>              | 474.3248   | 474.1922   | 632.0404 | 2.482                                             | 3.900e+07            | –              | –          | 474 nm                                                |
| 5s5p <sup>3</sup> P <sub>0</sub> – 5s5d <sup>3</sup> D <sub>1</sub>                         | 483.3393   | 483.2042   | 620.2526 | 2.349                                             | 3.300e+07            | –              | –          | 483 nm repumper                                       |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s5d <sup>3</sup> D <sub>3</sub>                         | 496.3648   | 496.2263   | 603.9761 | 5.093                                             | 6.140e+07            | –              | –          | 496.36 nm                                             |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s5d <sup>3</sup> D <sub>2</sub>                         | 496.9328   | 496.7942   | 603.2857 | 1.969                                             | 1.280e+07            | –              | –          | 496.93 nm repumper                                    |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s5d <sup>3</sup> D <sub>1</sub>                         | 497.3055   | 497.1668   | 602.8336 | 0.487                                             | 1.300e+06            | –              | –          | 497.30 nm                                             |
| 5s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s5p <sup>3</sup> P <sub>2</sub>              | 671.2065   | 671.0212   | 446.6471 | –                                                 | 1.520e-04            | 3.183e-10      | 7.6        | 671 nm magnetic-quadrupole line                       |
| 5s5p <sup>3</sup> P <sub>0</sub> – 5s6s <sup>3</sup> S <sub>1</sub>                         | 679.2894   | 679.1020   | 441.3324 | 1.970                                             | 8.360e+06            | 11.43          | 11.6       | 679 nm repumper                                       |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6s <sup>3</sup> S <sub>1</sub>                         | 688.0211   | 687.8313   | 435.7315 | 3.437                                             | 2.450e+07            | 11.43          | 34.1       | 688 nm                                                |
| 5s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s5p <sup>3</sup> P <sub>1</sub> <sup>o</sup> | 689.4491   | 689.2589   | 434.8290 | 0.151                                             | 4.700e+04            | 0.007479       | 100        | 689 nm red MOT                                        |
| 5s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s5p <sup>3</sup> P <sub>0</sub>              | 698.4456   | 698.2530   | 429.2281 | –                                                 | –                    | 1.349e-09      | –          | 698 nm optical clock / optical qubit                  |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6s <sup>3</sup> S <sub>1</sub>                         | 707.2022   | 707.0072   | 423.9134 | 4.525                                             | 3.910e+07            | 11.43          | 54.3       | 707 nm repumper                                       |
| 5s5p <sup>3</sup> P <sub>1</sub> – 5s4d <sup>1</sup> D <sub>2</sub>                         | 1771.3690  | 1770.8854  | 169.2434 | 0.164                                             | 1.950e+03            | 0.0003772      | 82         | 1D2 decay returning atoms to the ground state via 3P1 |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s4d <sup>1</sup> D <sub>2</sub>                         | 1904.3484  | 1903.8286  | 157.4252 | 0.084                                             | 4.180e+02            | 0.0003772      | 17.7       | 1D2 decay to the metastable reservoir state           |

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.

Γ/2π: natural linewidth of the upper level, 1/(2πτ).

## Isotope shifts

| Transition                                                                                  | λ vac (nm) | Isotopes | Shift (MHz) |
|---------------------------------------------------------------------------------------------|------------|----------|-------------|
| 5s5p <sup>3</sup> P <sub>0</sub> – 5s6d <sup>3</sup> D <sub>1</sub>                         | 394.192    | 88-84    | 91.05(4)    |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>1</sub>                         | 394.192    | 88-86    | 54.60(3)    |
| 5s5p <sup>3</sup> P <sub>0</sub> – 5s6d <sup>3</sup> D <sub>1</sub>                         | 394.192    | 88-87    | 51.641(28)  |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>3</sub>                         | 403.152    | 88-84    | 85(10)      |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>3</sub>                         | 403.152    | 88-86    | 41(10)      |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>3</sub>                         | 403.152    | 88-87    | 53(5)       |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>2</sub>                         | 403.352    | 88-84    | 81.0(20)    |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>2</sub>                         | 403.352    | 88-86    | 49.0(20)    |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>2</sub>                         | 403.352    | 88-87    | 20(5)       |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>1</sub>                         | 403.433    | 88-84    | 77(10)      |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>1</sub>                         | 403.433    | 88-86    | 30(10)      |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s6d <sup>3</sup> D <sub>1</sub>                         | 403.433    | 88-87    | 18.0(20)    |
| 5s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s5p <sup>1</sup> P <sub>1</sub> <sup>o</sup> | 460.862    | 84-88    | -275(3)     |
| 5s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s5p <sup>3</sup> P <sub>1</sub> <sup>o</sup> | 460.862    | 86-88    | -126.0(20)  |
| 5s <sup>2</sup> <sup>1</sup> S <sub>0</sub> – 5s5p <sup>3</sup> P <sub>1</sub> <sup>o</sup> | 460.862    | 87-88    | -47.0(20)   |
| 5s5p <sup>3</sup> P <sub>0</sub> – 5s5d <sup>3</sup> D <sub>1</sub>                         | 483.339    | 88-84    | 90.00       |
| 5s5p <sup>3</sup> P <sub>0</sub> – 5s5d <sup>3</sup> D <sub>1</sub>                         | 483.339    | 88-86    | 40.00       |
| 5s5p <sup>3</sup> P <sub>0</sub> – 5s5d <sup>3</sup> D <sub>1</sub>                         | 483.339    | 88-87    | 32.00       |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s5d <sup>3</sup> D <sub>3</sub>                         | 496.365    | 88-84    | 91.60(20)   |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s5d <sup>3</sup> D <sub>3</sub>                         | 496.365    | 88-86    | 46.60(20)   |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s5d <sup>3</sup> D <sub>3</sub>                         | 496.365    | 88-87    | 27.0(20)    |
| 5s5p <sup>3</sup> P <sub>2</sub> – 5s5d <sup>3</sup> D <sub>2</sub>                         | 496.933    | 88-84    | 92.80(20)   |

Shift = ν(first isotope) – ν(second isotope).